

A Unified Framework for Understanding Confinement Loss, and Engineering Bandwidth, and Modal-Dependent Loss in Hollow-Core Fibers

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Abstract Exploiting a unified framework linking mode-coupling and tube-tunneling losses in Hollow-Core Inhibited-Coupling Fibers, we show how tube shape and position govern transmission bandwidth and anisotropic leakage, revealing Mode-Dependent loss in HC-IC fibers as a controllable outcome of geometric and azimuthal design choices. ©2026 The Author(s)

Introduction

Hollow-core photonic crystal fibers based on the inhibited-coupling (IC) mechanism have broken the Rayleigh-scattering transmission limit of solid-core glass fibers and, thanks to the progress in the fabrication techniques, are now the leading platform for broadband, with ultralow loss and latency [1,2]. As a result of the technological advancement, the measured losses are reaching the limit set by the Confinement Loss (CL). Exploiting a recently presented unified framework [3], which reconcile the Antiresonance model [4] and the Inhibited Coupling one [5], we address three open challenges in hollow-core fiber deployment: expanding the usable C+L band window, controlling differential mode attenuation for mode-division multiplexing (MDM), and understanding polarisation-dependent loss in few-mode transmission.

Disentangling MCL and TTL

Numerical finite-element calculations for Tube Lattice Fibers (TLF) (Fig.1a), nested (N), double-nested (DN), and triple-nested (TN) fibers reveal

a universal spectral signature within each transmission band: Following the solid lines in Fig.1b is possible to follow the CL decreasing sharply from high wavelength λ , then flattening to a plateau before increasing again. Fig. 1b shows that the plateau level decreases with increasing number of nested elements in the cladding tubes, while the decreasing red front of the transmission band is fixed for all the considered fibers which are characterized by the same tube thickness t , core diameter D_{co} , widest tube diameter d_{ext} and number of cladding elements. In [3] we disentangled the two loss mechanisms drawing this characteristic loss spectrum shape. The two mechanisms are the Mode coupling Loss (MCL), which arises from coupling between the fundamental core mode (FCM) and dielectric modes (DMs) confined in the tube membranes, so it cannot be suppressed by adding nested tubes, a technique that is effective only against the second mechanism called Tube Tunneling Loss (TTL), due to the power leakage through the cladding tubes. In particular TTL for TLF goes as:

$$TTL(\lambda) = F_{cc}(\lambda) \cdot CL_{min}/F_{cc,min} \quad (1)$$

where F_{cc} is the normalised interface field

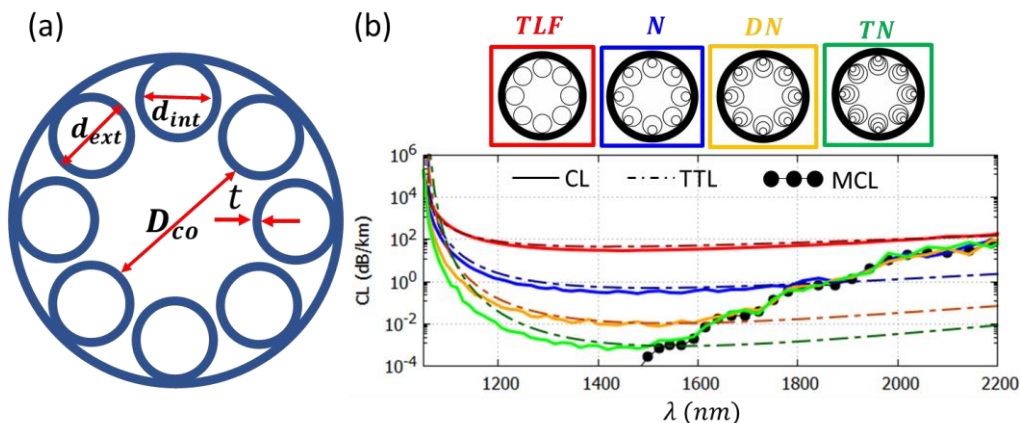


Fig. 1: (a) Fiber cross section. (b) Confinement Loss decreases with increasing number of nested elements (solid lines) as does the TTL (dashed lines); while MCL (dotted line) is common for all the considered fibers. It does not depend on the nested order.

integral [7]. For nested fibers with N_s nested layers, TTL scales as:

$$1 / \left(\sin \left(\frac{2\pi t_{eff}}{\lambda} \right) \right)^{2N_s}, \quad (2)$$

where $t_{eff} = 2t(\sqrt{nd^2 - 1})$ and nd silica refractive index. The sum MCL + TTL reproduces full numerical CL spectra, establishing for the first time a unified semi-analytical model of IC-HCF confinement loss.

Bandwidth Engineering via Tube Geometry

Within each transmission band, the spectral region dominated by MCL directly limits the usable bandwidth of IC-HCFs fixing both the red edge and the blue edge. Since the introduction of nested elements does not influence MCL, reducing it requires acting on the shape of the cladding elements themselves, which controls both the overlap integral between the FCM and DM fields and the spectral distribution of DM cut-off wavelengths [5]. We propose replacing the standard circular tube cross-section with an elliptical one, characterised by axial ratio b/a . Using the technique developed in [3] is possible to isolate the MCL contribution, that does not depend neither by the “length” of the tube which increases when the b axes of the ellipse increases. Fig. 2 shows that augmenting b/a from 1 (circular) to 1.4 progressively red-shifts the entire MCL spectrum, pushing it toward longer wavelengths and thereby enlarging the low-loss window reaching the maximum enlargement of 220nm at $b/a = 1.3$, after which the trend reverses. To assess the benefit under realistic conditions, in Fig. 3 we compare two fibers with the same core diameter ($D_{co} = 36 \mu\text{m}$), same number of tubes ($N_t = 8$), same wall thickness ($t = 0.5 \mu\text{m}$), and matched minimum CL: a double-nested fiber (DNF) with circular tubes and a single-nested fiber (NF) with elliptical tubes at $b/a = 1.3$. Matching minimum CL, rather than nesting order, provides the fairest performance comparison. The two CL spectra overlap for wavelengths shorter than $\sim 1400 \text{ nm}$, confirming that TTL governs this range equally for both geometries. Beyond 1400 nm , where MCL takes over and sets the red edge of the transmission band, the elliptical fiber shows markedly lower loss. For a more realistic evaluation, the Total Loss (TL), the sum of CL, Surface Scattering Loss (SSL), and Micro-Bending Loss (MBL) [8], is computed for both fibers. SSL and MBL become negligible at long wavelengths, so TL converges to CL and ultimately to MCL beyond $\sim 1600 \text{ nm}$ (circular) and $\sim 1700 \text{ nm}$ (elliptical). Taking the current minimum loss of silica solid-core fiber (0.14 dB/km) as a benchmark, the elliptical design extends the sub- 0.14 dB/km bandwidth by $\sim 150 \text{ nm}$, corresponding to a 30% bandwidth gain relative to the circular.

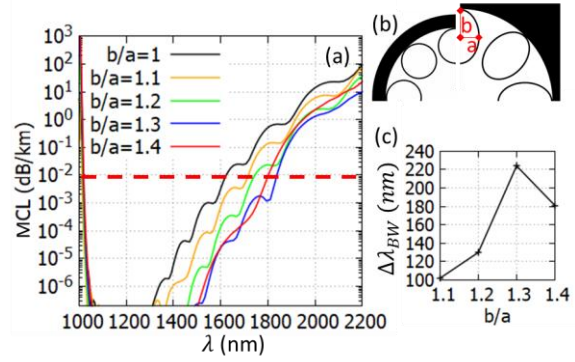


Fig. 2: (a) MCL of fibers with tubes with different axial ratios. (b) Cross sections of TLF with circular and elliptical tubes. (c) Bandwidth enlargement $\Delta\lambda_{BW}$ at MCL level of 10^{-2} dB/km with respect to circular tubes fiber.

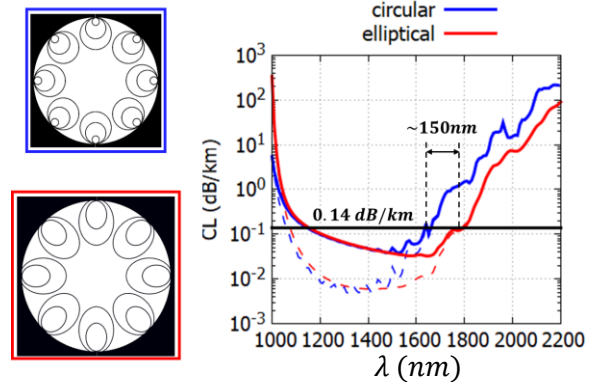


Fig. 3: Left: Cross sections of the DN circular elements (blue) and N elliptical elements with aspect ratio 1.3 (red) fibers. Right: CL (dashed) and TL (solid) spectra of the two fibers.

TTL Anisotropy as a Lever for Modal-Dependent Loss Engineering

Mode-dependent loss (MDL) is a fundamental impairment in mode-division multiplexing systems, as unequal modal attenuation limits capacity and imposes severe constraints on MIMO equalisation complexity [9]. A key finding of this work is that TTL is intrinsically anisotropic, contrary to the common assumption of isotropic leakage. Radial Poynting-vector maps at the jacket boundary reported in Fig. 4 reveal that the leakage intensity through each individual tube depends critically on how the modal electric field is oriented relative to that tube's boundary. For x and y-polarised FM components, the field maxima face different tubes at different azimuthal positions, producing unequal TTL contributions per tube and a net polarisation-dependent confinement even in otherwise geometrically symmetric claddings. This same TTL anisotropy is the primary driver of MDL in IC-HCFs. Extending the MCL/TTL framework to higher-order mode groups LP01, LP11, and LP21, we find that TTL, governing the low-loss plateau, is the key discriminator: higher-order modes, especially LP21, exhibit stronger field

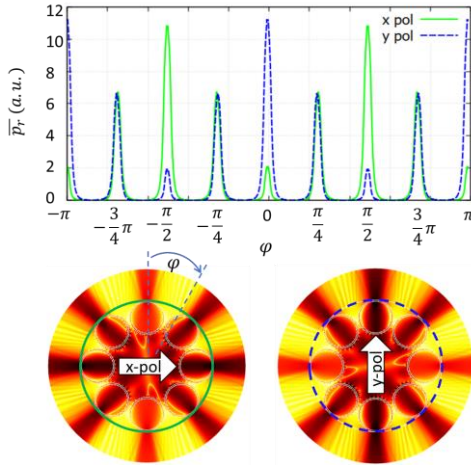


Fig. 4: (up) Radial Poynting (p_r) vector recorded at the jacket interface. It is visible different leakage distribution through the tubes between x and y polarization. (down) p_r vector map of Fundamental Mode y- and x-polarized.

penetration toward the cladding and a different azimuthal field distribution, causing their field maxima to face individual tube boundaries more directly and thus experience substantially higher TTL-driven leakage. Because TTL depends on both the azimuthal field distribution and the polarisation of each mode, different sub-classes within LP11 and LP21 experience markedly different leakage intensities through each tube as it is visible from the CL of a fiber with 8 uniform nested tubes in Fig. 5b. This polarisation and mode-selective nature of TTL is not merely a limitation, but a design degree of freedom. We explore a tailored cladding architecture employing a mix of 12 tubes single- and double-nested with controlled azimuthal placement to deliberately manipulate TTL pathways. By adjusting the TTL of individual

tubes through their azimuthal position and Ns number, it is possible to reorder and partially equalise modal losses across the LP01, LP11, and LP21 families. Radial Poynting-vector reported in Fig.5a confirms that leakage channels can be engineered to shift modal ordering and compress the MDL spread, demonstrating that MDL in IC-HCFs is a controllable design parameter rather than an intrinsic constraint. Thus, by compressing the MDL spread across LP₀₁, LP₁₁, and LP₂₁, this approach relaxes the MIMO processing burden and opens a concrete path toward few-mode hollow-core links.

Conclusions

In conclusion, this work exploits a unified semi-analytical model to understand CL, and engineering bandwidth, and MDL in Hollow-Core Fibers. The model successfully disentangle mode-coupling loss (MCL) and tube-tunneling loss (TTL). In this framework, we demonstrate that the red-edge bandwidth is not a fixed limit but a controllable parameter: replacing standard circular tubes with elliptical geometries ($b/a = 1.3$) yields a 30% enlargement in the low-loss window. Furthermore, the model reveals that TTL is intrinsically anisotropic, proving that Mode-Dependent Loss (MDL) in nested structures is a direct consequence of the interaction between modal field orientation and azimuthal cladding symmetry. Ultimately, this framework provides the physical insight necessary to transform hollow-core fiber design from empirical trial-and-error into a deterministic process for engineering ultralow loss and controlled multi-mode operation.

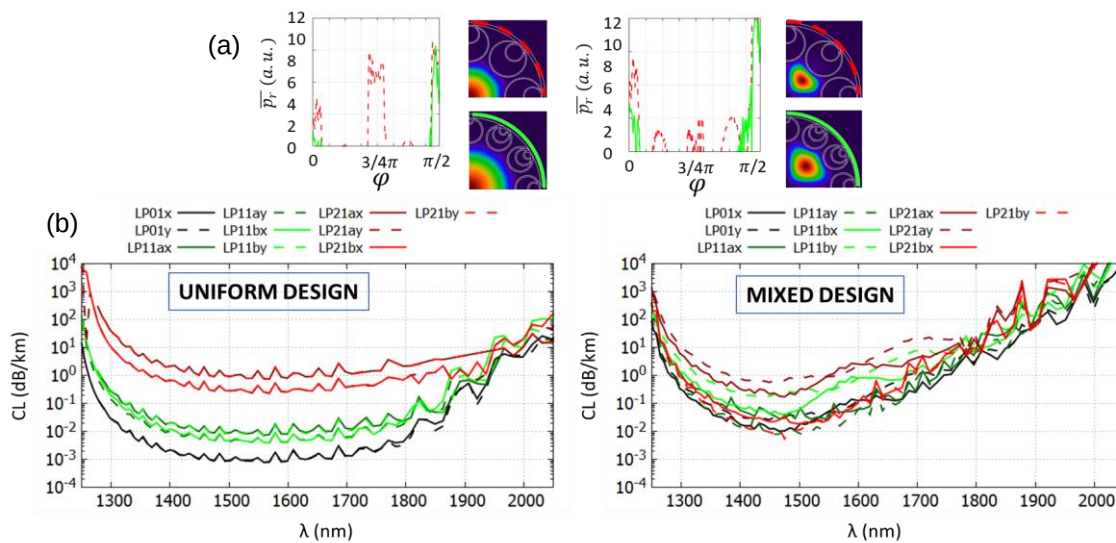


Fig. 5: (a) Radial Poynting (p_r) vector recorded at the jacket interface of the FM x-polarized and LP11 x-polarized in case uniform and mixed cladding design. The uniform design is characterized by 8 single nested tubes, mixed design by 12 tubes with 8 double nested and 4 single nested on the vertical and horizontal symmetry axes of the fiber. (b) Comparison between the CL of LP01, LP11 and LP21 modes for the two different cladding designs.

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